

Total No. of Questions : 8]

SEAT No. :

P-7631

[Total No. of Pages : 2

[6180]-151

T.E. (Information Technology)

SOFTWARE MODELING & DESIGN

(2019 Pattern) (Semester - II) (314454D) (Elective - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q. 8.*
- 2) *Assume suitable data if required.*
- 3) *Figures to the right indicates full marks.*

Q1) a) Analyze the scenario for an end-semester examination form registration process and draw activity diagram using the concept of fork-join, constraints and swim-lanes. [8]

b) Draw the notations of synchronous asynchronous, return, create object and destroy object messages in the context of sequence diagram. [9]

OR

Q2) a) Analyze the scenario for online purchase process from an e-commerce website and draw a sequence diagram using the concepts of object life line and message. [8]

b) In the context of a state diagram, explain concurrent states, nested states, sub-machine states and composite states with suitable examples. [9]

Q3) a) Explain steps in designing business classes. [6]

b) Justify the use of activity diagram in design of business class. [6]

c) What are the benefits of the Access Layer Class? List the access layer tasks. [6]

OR

Q4) a) What is OCL? Explain its various mechanisms and its applications.[6]

b) Describe the steps involved in the process of designing view layer.[6]

c) Draw a detailed Deployment diagram for Hotel management system scenario. [6]

P.T.O.

Q5) Explain following design patterns with suitable application design requirements : **[18]**

- a) Factory design pattern
- b) State design pattern
- c) Adapter design pattern

OR

Q6) a) List GOF patterns. Explain any three patterns in detail. **[9]**

b) Elaborate on General Responsibility Assignment Software Patterns. **[9]**

Q7) a) Explain the entities involved in documenting a software architectural pattern. **[8]**

b) What is service oriented architecture? Explain its applications. **[9]**

OR

Q8) a) List and explain multiple views of a software architecture. **[9]**

b) Write a short note on: product line software architectures. **[8]**

